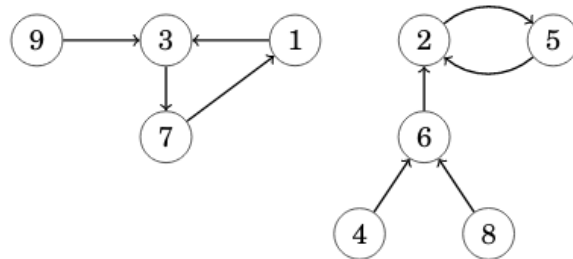


Functional Graph Distribution

Time limit: 1.00 s

Memory limit: 512 MB

A *functional graph* is a directed graph where each node has outdegree 1. For example, here is a functional graph that has 9 nodes and 2 components:



Given n , your task is to calculate for each $k = 1 \dots n$ the number of functional graphs that have n nodes and k components.

Input

The only input line has an integer n : the number of nodes.

Output

Print n lines: for each $k = 1 \dots n$ the number of graphs modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 5000$

Example

Input	Output
3	17 9 1